

WEB 268 Final Project Requirements

In the web capstone course, you will be required to develop a **web application**, not a website. This application is not required to be an application for a non-profit business, as in the past. It should be remembered however that this application is meant as a portfolio builder to be included in each team member's portfolio, when searching for employment. This final project will be developed using current tools and techniques consistent with the current web industry. Look at the requirements listed below.

Your WEB application will also include multimedia titles developed from scratch. This will include images, audio, and video elements on your applications web pages. You will demonstrate proficiency from the design courses of the WEB program.

You will use appropriate design tools and demonstrate a variety of skills learned in the WEB design courses.

The application will consist of both a front-end and a back-end. The application must be a web app, i.e., no desktop apps will be allowed. You may develop a mobile app for either Android or iPhone if you choose.

Your web application will be developed for either a real or a fictitious business.

General Requirements

The students will be organized into teams. I would like three to four members per team. This should be done by the second week of class. The team is required to develop all documentation consistent with a real-world web application. This includes a proposal, ERD chart, storyboard, data flow diagrams, status reports (bi-weekly). By the last week of class each team will give a presentation of their application.

In the presentation you will show:

1. Tools used
2. Components the Team members designed. (must be at least five)
3. Design titles will include: Audios, videos, and images. Must have some of each.
4. A backend with a database and server software
5. A frontend with screen definitions in HTML or some sort of view engine
6. The database will include at least four data tables with at least ten fields each
7. The app will perform CRUD operations on the data tables
8. All pages/screens will be worthy of a professional WEB application

As of the Fall of 2025, AI is becoming prevalent in WEB applications. AI and Workflow based system is being incorporated into the WEB program at HACC at this time. However, I will encourage you to include AI in your WEB app as possible. This included incorporated Chatbots, web links to workflows, any and all AI assisted processes in your web app. Note: this is not required at this time, but will certainly improve your marketability in the industry. The reason will be explained by the instructor during the course.

Each team member will contribute part of the status report and presentation. Each team member is required to code some parts of the WEB application. I do not want one person doing the whole project. Any complaints about a non-participating team member may result in points lost for the final grade.

The bi-weekly status reports will include what was done in the WEB application.

The team will use an appropriate web application development framework from the following list:

1. Python and Django
2. Javascript with React and/or Angular (I would not use both) a NodeJS app
3. ASP.NET and C# (No Visual Basic)
4. PHP with CakePHP or Laravel
5. Straight NodeJS application with Node frameworks
6. Wordpress

Allowable DBMS (Do NOT use SQLite)

1. MySQL
2. MongoDB
3. SQL Server

You will have at least four tables or collections in your app. No less than three of those tables must be CRUD targets from the front-end. Each table must have at least ten fields. This does not count system tables for things like valid zip-codes and state-names. Of course, systems tables would be used for validation in most cases. For example, valid state names.

You will be allowed to use Wordpress or any other CMS. You must use technologies taught in the WEB Program. If you use a CMS, a database must be involved and used. I prefer you use custom tables. If you choose to use a CMS, such as Wordpress, you may use plugins and themes. However, you must do significant coding and design work worthy of Capstone. I (the instructor) will ultimately decide what is worthy coding and design. You will NOT use just plugins and themes to write an app, you must code and design. If you do use Wordpress, I strongly recommend you use the Elementor plugin to achieve really significant page designs.

We have not used AI or RAG in the WEB program to date. However, if you incorporate AI in your applications I will award extra points to the entire team. The App should use AI agents to assist the user of the application in some way. Because we have not incorporated AI or RAG in the WEB program to date, I will leave the requirements open now. But I do expect to see a clear use of either OpenAI/ChatGPT or a local LLM in the app. This requirement must be clearly demonstrated in the final project presentation. The instructor will determine based on the usage of AI, how many additional points are awarded.

Front End Requirements

1. HTML
2. Generated HTML from a view engine such as Razor
3. Proper use of CSS (I strongly recommend using Bootstrap or other CSS framework)

Backend Requirements

1. You may use graphql for data packaging

Design Requirements

The WEB application will use multimedia titles developed by the team. This included Audio, Videos, and images. The project will require ample design and CSS elements.

Documentation Library

It is important to remember, before my life at HACCC I was a real developer. So, I know what works!!!! No one keeps all the information in their heads. To that end you will be required to build for yourselves a library of documentation, which includes: PDF or actual books. Videos, Working applications. I recommend all class members collect books on all computer technologies and share. I can tell when I was developing systems I had up to a dozen books open at any one time. To be successful you will never stop learning, so you need to during your working careers to maintain a library of books and code to save time and effort. This will be part of your grade.

Deliverables

1. System proposal
2. Storyboards for screen flow
3. ERD charts for database diagramming
4. Well-commented source code
5. Screen prints in the story board

Presentation

At the end of the course each team (*all members*) will present their application to the class and instructor. Each member will describe:

1. their role in the team
2. what parts of the application they developed

Recommendations

After your proposal, develop the database first before you start coding your app. Get it in a semi-completed form before write the application proper. Remember, even a small database change can cause a lot of app re-coding.

Get a book on MEAN or MERN or ASP.NET development that creates an app and follow it when you are developing your own app. I recommend Pro MERN stack development 2nd edition from apress.

Another recommendation is to search for a Udemy course on the technology you want to use and follow it for help. Many Udemy courses develop systems worthy of Capstone.